

# **Application and extensions Meta-analysis of diagnostic studies**



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# Purpose of a diagnostic study

- Evaluate the quality of diagnostic test: does the test discriminate between positive and negative cases?
- Compare the quality of multiple diagnostic tests



# Quality of diagnostic test?

The « truth» ('golden standard')

|                               |          | Positive                  | Negative                  |
|-------------------------------|----------|---------------------------|---------------------------|
| Decision based<br>on new test | Positive | <b>True<br/>positive</b>  | <b>False<br/>positive</b> |
|                               | Negative | <b>False<br/>negative</b> | <b>True<br/>negative</b>  |



# Example:

## Screening instrument for Down Syndrome

The « truth» ('golden standard')

|                               |          | Positive | Negative     |
|-------------------------------|----------|----------|--------------|
| Decision based<br>on new test | Positive | <b>9</b> | <b>351</b>   |
|                               | Negative | <b>1</b> | <b>4,449</b> |



# A measure of the quality: DOR

- The (diagnostic) odds ratio (DOR):

$$\frac{TP * TN}{FP * FN} = \frac{9 * 4449}{351 * 1} = 114.08$$

- Log odds ratio:  $\ln_e(114.08) = 4.74$
- $\sigma_{LOR}^2 = \frac{1}{9} + \frac{1}{351} + \frac{1}{1} + \frac{1}{4449} = 1.11$
- $CI(LOR) = [4.74 - 1.96 * \sqrt{1.11}; 4.74 + 1.96 * \sqrt{1.11}]$   
 $= [2.67; 6.81]$

- Remark: continuity  
correction: add .5 to all cells

Positive

Negative

Positive

Negative

|   |       |
|---|-------|
| 9 | 351   |
| 1 | 4,449 |

# Meta-analyses of DORs

- Ordinary precision-weighted meta-analysis!

- Note:  $g \cong \frac{\sqrt{3}}{\pi} * \text{LOR} = 2.61$

$$\sigma_g^2 \cong \frac{3}{\pi^2} * \sigma_{\text{LOR}}^2 = 0.34$$

⇒ Possible to convert  $g \leftrightarrow \text{LOR}$  and combine in one analysis

- But: loss of information:  
different tables can give the same DOR



# Measures of quality: $s_n$ and $s_p$

- The sensitivity ( $s_n$ ):  $TP/P = 9/10 = 90\%$   
(probability that a positive case will be tested positively)
- The specificity ( $s_p$ ):  $TN/N = 4449 / (351 + 4449) = 93\%$   
(probability that a negative case will be tested negatively)
- Note:
  - False Positive Rate (FPR) =  $1 - s_p = 351 / (351 + 4449) = 7\%$
  - DOR =  $LR_+ / LR_-$

$$= (s_n / (1 - s_p)) / (1 - s_n / s_p)$$

Positive

Negative

|          | <u>Positive</u> | <u>Negative</u> |
|----------|-----------------|-----------------|
| Positive | 9               | 351             |
| Negative | 1               | 4,449           |

# Meta-analysis of $s_n$ and $s_p$

Juneyoung, Kyung Won, Sang Hyun, Jimi, & Seong Ho (2015).

Separate meta-analyses of  $s_n$  and  $s_p$   
(precision-weighted meta-analyses)

Bivariate (random effects) meta-analysis:

**Level 1:** # of positive test results for positive cases in study  $i$

$$\begin{cases} Y_{i|positive} \sim \text{Bin}(n_{1i}, \pi_{1i}) \\ Y_{i|negative} \sim \text{Bin}(n_{2i}, \pi_{2i}) \end{cases}$$

Number of positive cases

$\mathbf{S}_n$

$1 - \mathbf{S}_p$  (FPR)

**Level 2:**

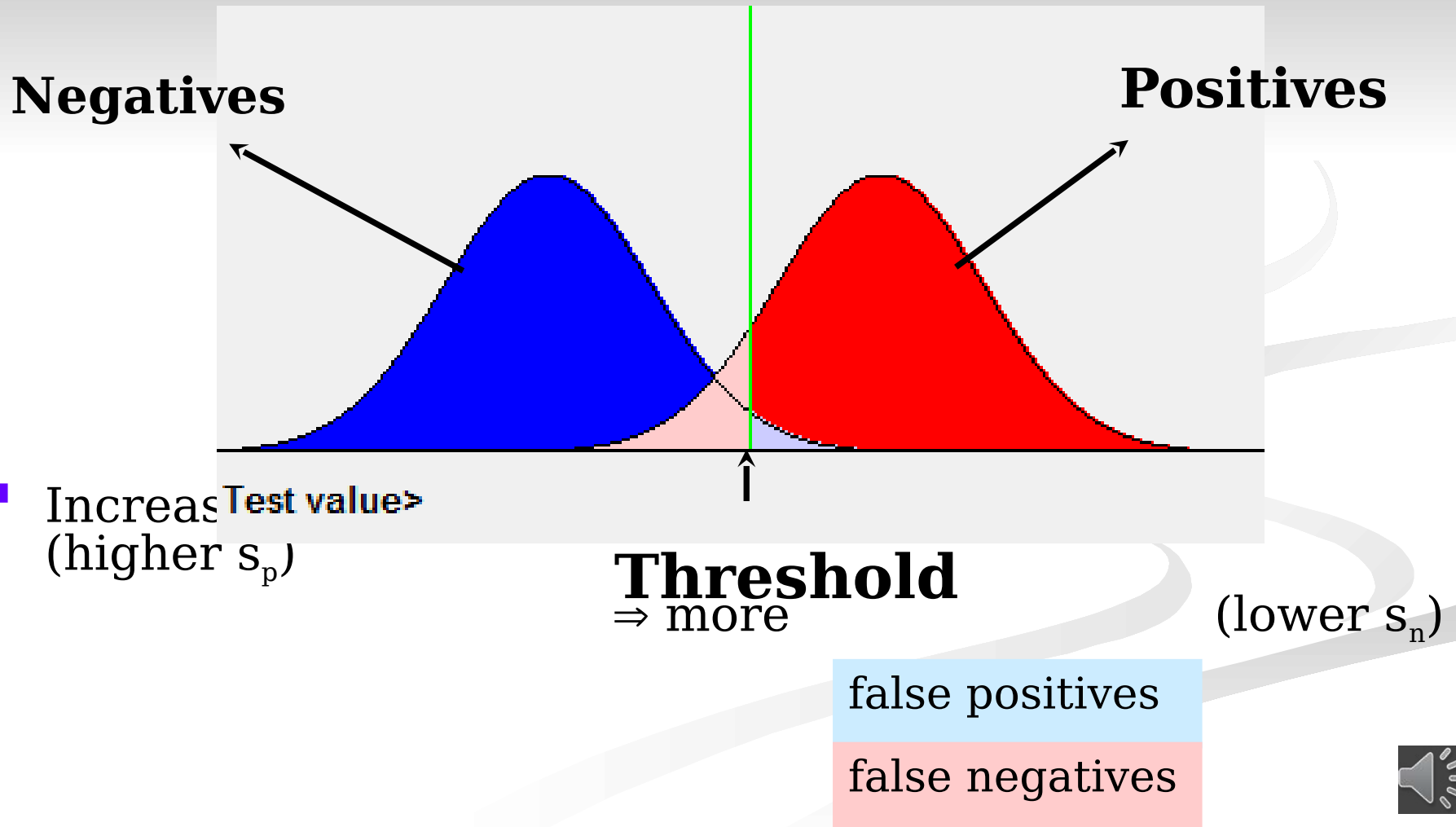
# of positive test results for negative cases in study  $i$

$$\begin{cases} \text{logit}(\pi_{1i}) = \theta_1 + u_{1i} \\ \text{logit}(\pi_{2i}) = \theta_2 + u_{2i} \end{cases} \text{ with } \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{u1}^2 & \sigma_{u1u2} \\ \sigma_{u1u2} & \sigma_{u2}^2 \end{bmatrix}\right)$$





- Note:  $s_n$  and  $s_p$  depend on the cut-off (or threshold) used!

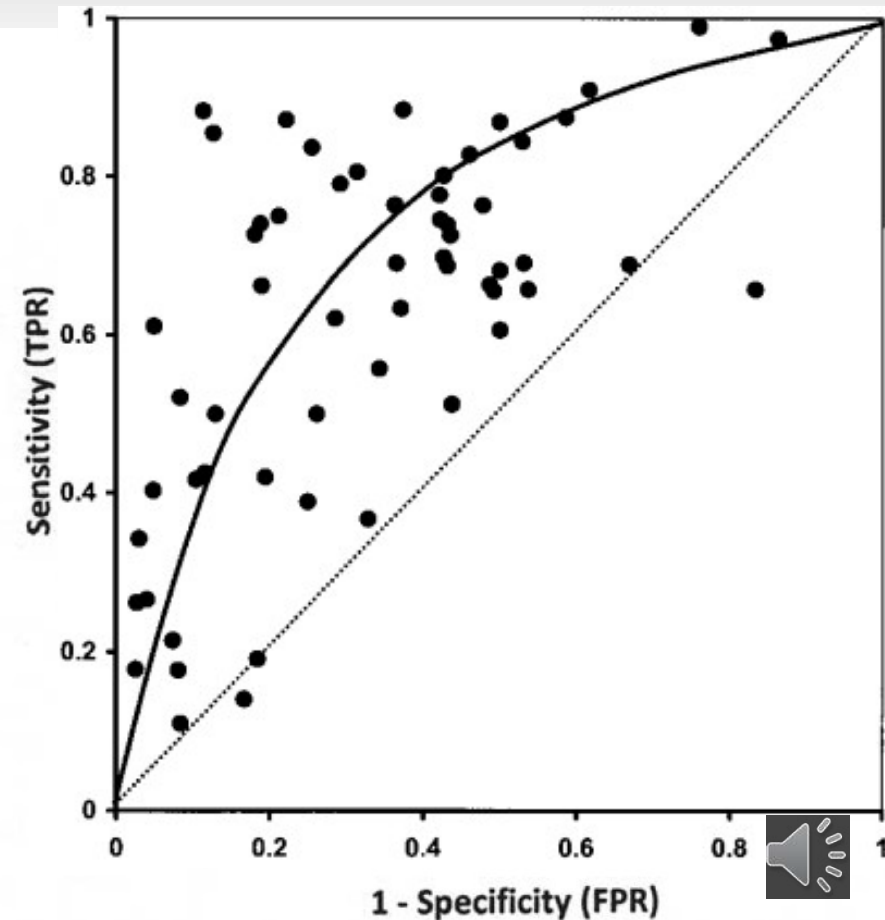


**$s_n$  and  $s_p$  depend on the cut-off used**

→ between-study variance in  $s_n$  or  $s_p$  also may reflect differences in threshold

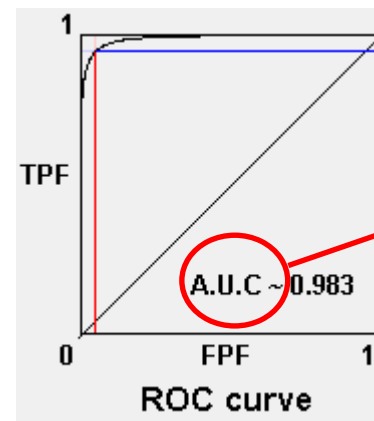
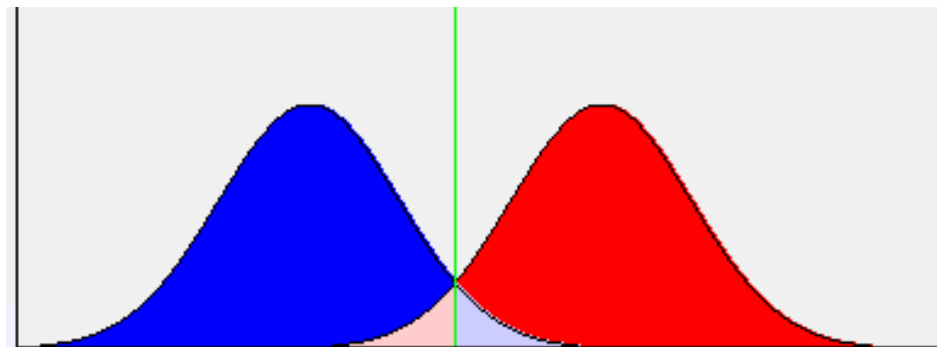
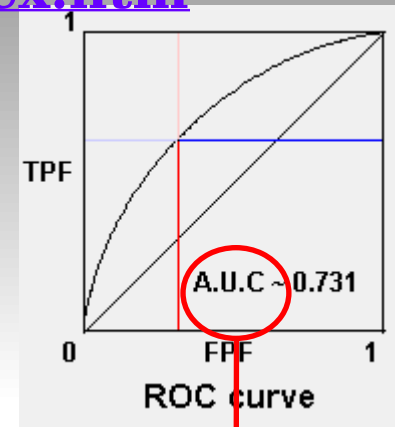
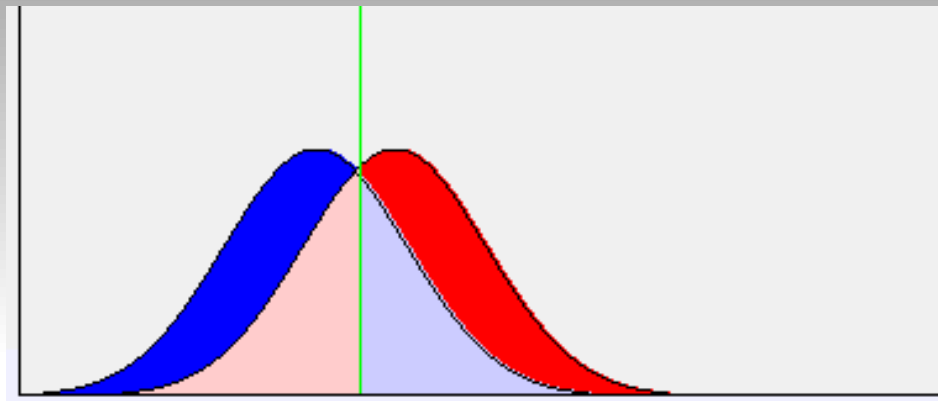
**Alternative:** (Summary) [Receiving Operating Characteristic-method](#)  
([SROC](#); summarizes data per study and/or over studies)

■ Increasing threshold  
⇒ more (lower  $s_n$ )



# The ROC-curve

<http://www.anaesthetist.com/mnm/stats/roc/Findex.htm>



**Area Under Curve**

- Find the best compromise in  $s_n$  and  $s_p$ , accounting for severity of disease, financial/emotional costs of both decisions, ...
- The higher AUC, the better!



# Meta-analysis of ROC-curves: The Hierarchical Summary ROC (HSROC)

## ■ Level 1:

$$\begin{cases} Y_{i|positive} \sim \text{Bin}(n_{1i}, \pi_{1i}) \\ Y_{i|negative} \sim \text{Bin}(n_{2i}, \pi_{2i}) \end{cases} \begin{matrix} \xrightarrow{\text{red circle}} \mathbf{S}_n \\ \xrightarrow{\text{red circle}} \mathbf{1} - \mathbf{S}_p \end{matrix}$$

## Level 2:

$$\text{logit}(\pi_{ji}) = (\theta_i + \alpha_i X_{ij}) \exp(-\beta X_{ij})$$

with  $j = 1$  for a positive subject, 2 for a negative subject

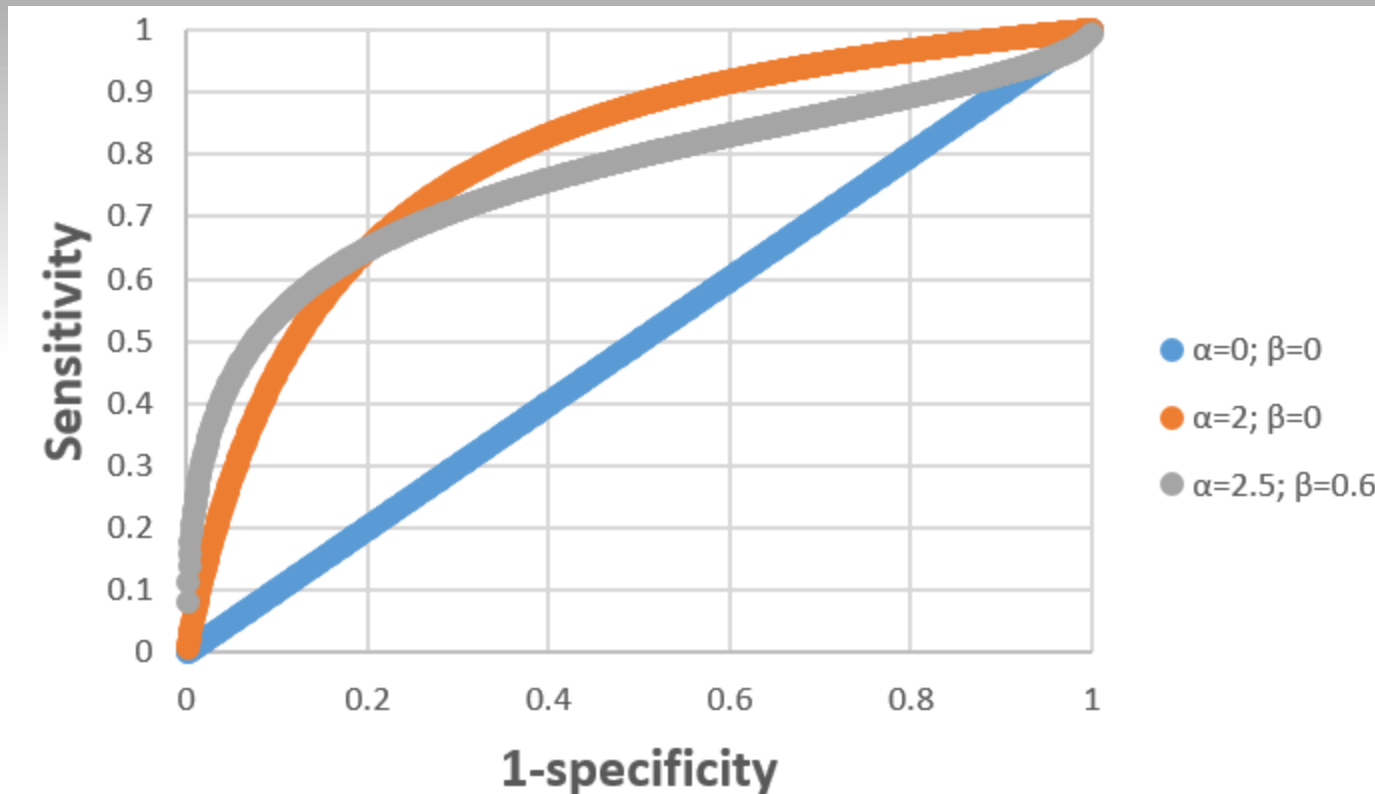
$X_{ij} = 1$  for a positive subject, 0 for a negative subject

$\theta_i$  and  $\alpha_i$  multivariate normally distributed over studies

$\Rightarrow$  Means and (co)variances of  $\theta_i$  and  $\alpha_i$  estimated.



$$\text{logit}(\pi_{ji}) = (\theta_i + \alpha_i X_{ij}) \exp(-\beta X_{ij})$$



- $\theta_i$ : determines the probability for a negative person of testing positively ( $\sim$ threshold used in study  $i$ )
- $\alpha_i$ : the higher, the better the test discriminates ( $\sim$ accuracy)
- $\beta$ : asymmetry of ROC

